

Bernoulli Expansion Proofs: Corrected Status

Plan 1 / Sharp stability for Faber–Krahn

2026-05-11

Abstract

This note corrects the overclaims in the previous Bernoulli expansion proof. The Bernoulli spectral idea remains promising, but the earlier writeup did not prove the required Schauder/tame remainder estimates. In particular, it incorrectly evaluated the interior harmonic first variation outside B_1 . The note below records the correct fixed-domain formulation and the conditional spectral closure that would follow from the missing tame estimates.

1 What Was Wrong

The previous proof defined

$$u_1(r, \theta) = \frac{1}{N} \sum_{k \geq 0} r^k g_k(\theta)$$

as the harmonic extension of g/N in B_1 , then evaluated it at $r = 1 + g(\theta)$. For a general $g \in C^{2,\gamma}$, this series is only an interior harmonic expansion. It has no controlled extension to $r > 1$. Thus the claimed estimates for

$$u_1((1 + g)\theta), \quad \partial_r u_1(1 + g, \theta)$$

are not valid.

Consequently, the claimed quadratic Schauder remainder

$$\|u_g - u_{B_1} - u_1\|_{C^{2,\gamma}} \lesssim \|g\|_{C^{2,\gamma}}^2$$

on U_g , and the derived tame estimate

$$\|R_q(g)\|_{L^2} \lesssim \|g\|_{C^{2,\gamma}} \|g\|_{L^2},$$

were not proved.

2 Correct Fixed-Domain Formulation

Let

$$\Phi_g(r\theta) = (r + \chi(r)g(\theta))\theta, \quad \tilde{u}_g = u_g \circ \Phi_g.$$

Then $\tilde{u}_g = 0$ on ∂B_1 , and

$$-\operatorname{div}(A_g \nabla \tilde{u}_g) = J_g \quad \text{in } B_1, \quad A_g = J_g D\Phi_g^{-1} D\Phi_g^{-T}.$$

This is the right framework. The first variation of the shape derivative u' still satisfies

$$\Delta u' = 0 \quad \text{in } B_1, \quad u'|_{\partial B_1} = \frac{g}{N}.$$

But the pullback derivative is not simply $u' \circ \Phi_g$ evaluated outside the ball. Boundary-gradient expansions must be derived on the fixed boundary from the pulled-back equation and the normal transformation formula.

3 The Useful Linearization

The first-order Bernoulli map is still expected to be

$$q_g = \frac{1}{N} - \frac{1}{N}(\mathcal{L}g) + \text{higher order}, \quad \mathcal{L}g_k = (k-1)g_k.$$

This part is consistent with standard shape calculus:

- u' has boundary data g/N ;
- the Dirichlet-to-Neumann map on the unit ball sends g_k to kg_k ;
- evaluating the normal derivative on the moved boundary contributes the additional $-g_k$, giving $k-1$.

The α -variation bracket also still has the linearization

$$\Psi_g = 1 + (I - \Pi_1)g + \text{higher order},$$

where Π_1 is the degree-one spherical-harmonic projection.

4 Conditional Spectral Closure

Proposition 1 (Conditional Bernoulli spectral closure). *Assume that in the small $C^{2,\gamma}$ graph class,*

$$q_g = \frac{1}{N} - \frac{1}{N}\mathcal{L}g + R_q(g), \quad \|R_q(g)\|_{L^2} \leq C\|g\|_{C^{2,\gamma}}\|g\|_{L^2},$$

and

$$\Psi_g = 1 + (I - \Pi_1)g + R_\alpha(g), \quad \|R_\alpha(g)\|_{L^2} \leq C\|g\|_{C^1}\|g\|_{L^2}.$$

Then, for selected minimizers satisfying

$$q_g^2 = \Lambda + a_\sigma \Psi_g, \quad |a_\sigma| \leq C\sigma,$$

small σ and small $\|g\|_{C^{2,\gamma}}$ force $g = 0$, after volume and barycenter normalization.

Proof. Squaring the assumed Bernoulli expansion and subtracting the constant part gives

$$-\frac{2}{N^2}\mathcal{L}g + R(g) = a_\sigma(I - \Pi_1)g + a_\sigma R_\alpha(g),$$

with

$$\|R(g)\|_{L^2} \leq C\|g\|_{C^{2,\gamma}}\|g\|_{L^2}.$$

Projecting to modes $k \geq 2$ gives

$$\mathcal{L}g_{\geq 2} = O(\sigma)g_{\geq 2} + O(\|g\|_{C^{2,\gamma}})g.$$

On this subspace $\|\mathcal{L}h\|_{L^2} \geq \|h\|_{L^2}$, so

$$\|g_{\geq 2}\|_{L^2} \leq C(\sigma + \|g\|_{C^{2,\gamma}})\|g\|_{L^2}.$$

Volume and barycenter normalization give

$$\|g_0\|_{L^2} + \|g_1\|_{L^2} \leq C\|g\|_{L^2}^2.$$

For small enough σ , $\|g\|_{C^{2,\gamma}}$, and $\|g\|_{L^2}$, this forces $g = 0$. □

5 What Remains Genuinely Open

To make the Bernoulli spectral route publication-ready, one must prove the tame Bernoulli expansion without evaluating the harmonic first variation outside B_1 . A viable approach is:

1. work entirely on B_1 with the pullback equation;
2. differentiate the coefficient map $g \mapsto (A_g, J_g, \nu_g)$;
3. prove a second-order fixed-domain remainder estimate for $\tilde{u}_g$ in a norm strong enough to control the boundary normal derivative;
4. show the boundary-gradient remainder is tame in the mixed form $C^{2,\gamma} \times L^2$, not merely $C^{2,\gamma}$ -quadratic.

Until that is done, the selection route should still be considered closed by BDV's nearly spherical second variation, not by the Bernoulli spectral closure.